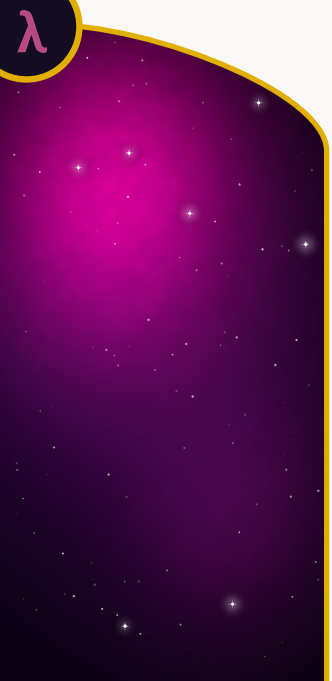


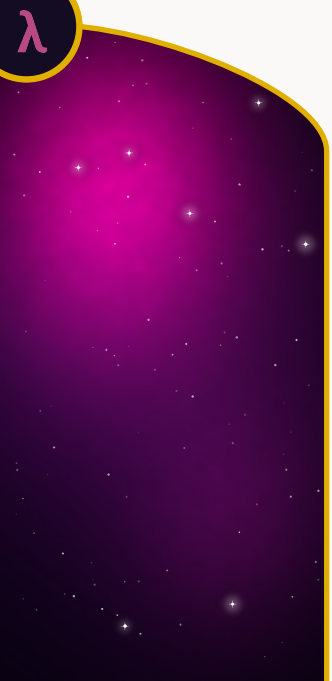
L12A: Metacircular Evaluator II

CS1101S: Programming Methodology

Martin Henz

4th November, 2026

Version 1.0 — Built: 2026-11-03 23:18:01 SGT



Recap: the evaluator so far

Applications and built-in functions

Adding functions

Local names

Running the evaluator

Lazy evaluation

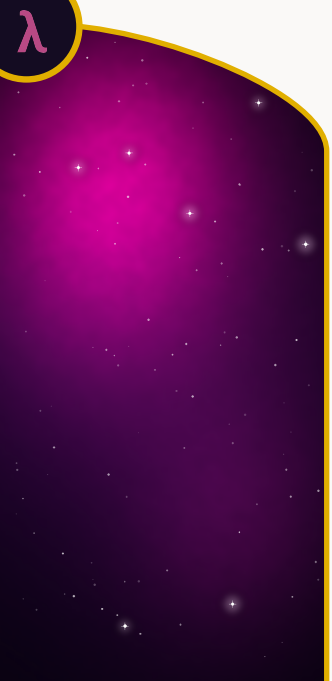
Recap: the evaluator so far

```
def evaluate(component, env):
    if is_literal(component):
        return literal_value(component)
    elif is_name(component):
        return lookup_symbol_value(
            symbol_of_name(component), env)
    elif is_operator_combination(component):
        return eval_operator_combination(component, env)
    elif is_conditional(component):
        return eval_conditional(component, env)
    elif is_sequence(component):
        return eval_sequence(
            sequence_statements(component), env)
    elif is_assignment(component):
        return eval_assignment(component, env)
    else:
        return error('unknown syntax', component)
```

Recap: the evaluator so far

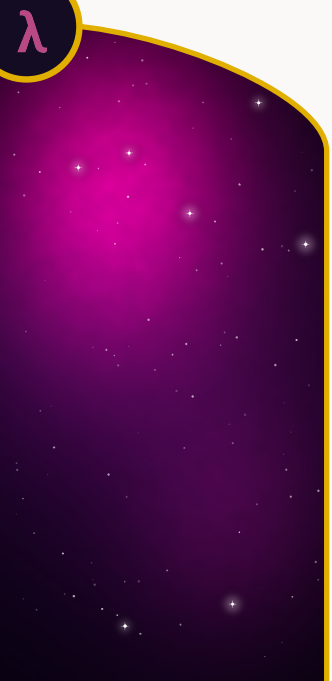
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    if is_literal(component):
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    elif is_operator_combination(component):
        return eval_operator_combination(component, env)
    elif is_conditional(component):
        return eval_conditional(component, env)
    elif is_sequence(component):
        return eval_sequence(
            sequence_statements(component), env)
    elif is_assignment(component):
        return eval_assignment(component, env)
    else:
        return error('unknown syntax', component)
```

- parse turns program string into syntax tree
- Environments are linked lists of frames



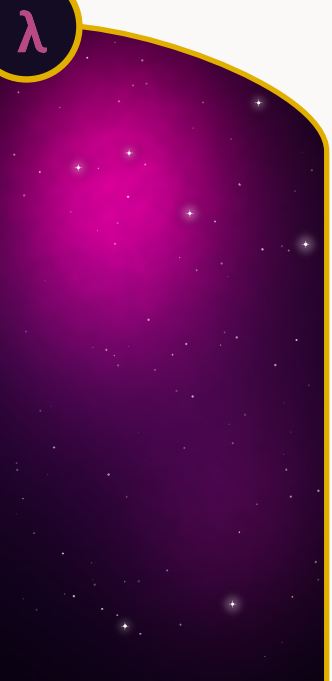
Today

- Applications and built-in functions: `evaluate` meets `apply`



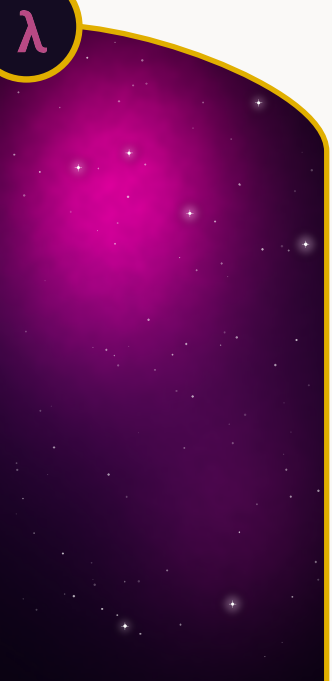
Today

- Applications and built-in functions: `evaluate` meets `apply`
- Adding functions: definitions, lambda expressions, applications and return statements



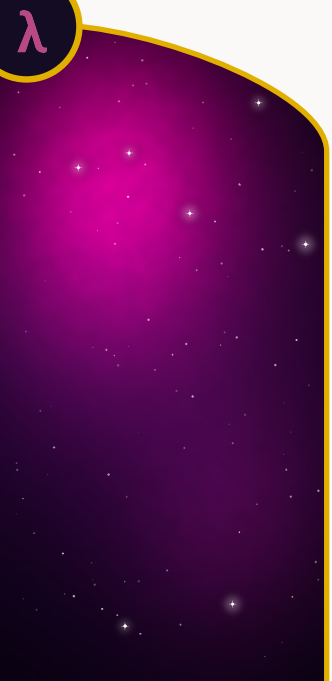
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- Applications and built-in functions: `evaluate` meets `apply`
- Adding functions: definitions, lambda expressions, applications and return statements
- Local names, `global` and `nonlocal`



Today

- Applications and built-in functions: `evaluate` meets `apply`
- Adding functions: definitions, lambda expressions, applications and return statements
- Local names, `global` and `nonlocal`
- Changing the language: lazy evaluation



Recap: the evaluator so far

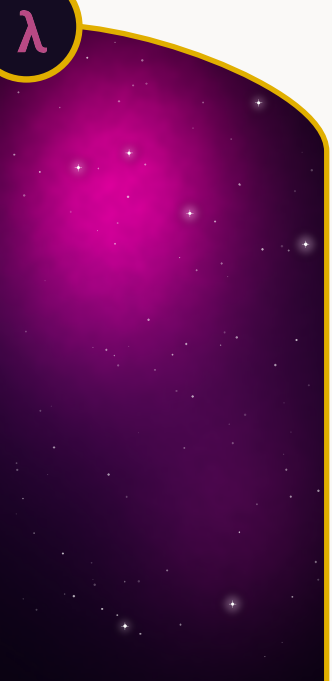
Applications and built-in functions

Adding functions

Local names

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Built-in functions in the evaluator

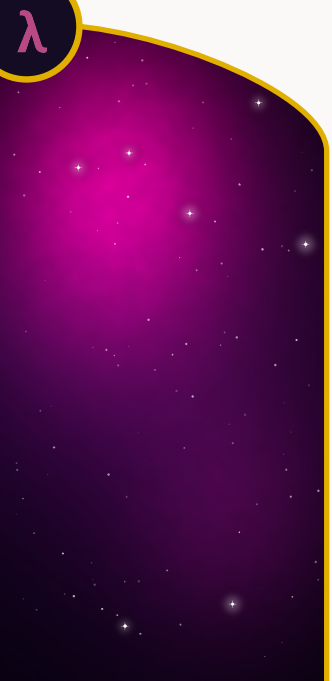
Built-in functions in the evaluator

```
built_in_functions = llist(
    llist('head', head),
    llist('tail', tail),
    llist('pair', pair),
    llist('is_none', is_none),
    llist('llist', llist),
    llist('print', print_without_environments),
    llist('error', error),
    llist('abs', abs),
    llist('+', lambda x, y: x + y),
    llist('-', lambda x, y: x - y),
    llist('*', lambda x, y: x * y),
```

Built-in functions in the evaluator

```
built_in_functions = llist(
    llist('head', head),
    llist('tail', tail),
    llist('pair', pair),
    llist('is_none', is_none),
    llist('llist', llist),
    llist('print', print_without_environments),
    llist('error', error),
    llist('abs', abs),
    llist('+', lambda x, y: x + y),
    llist('-', lambda x, y: x - y),
    llist('*', lambda x, y: x * y),
```

Built-in functions are linked lists that start with the string `'built_in'`.



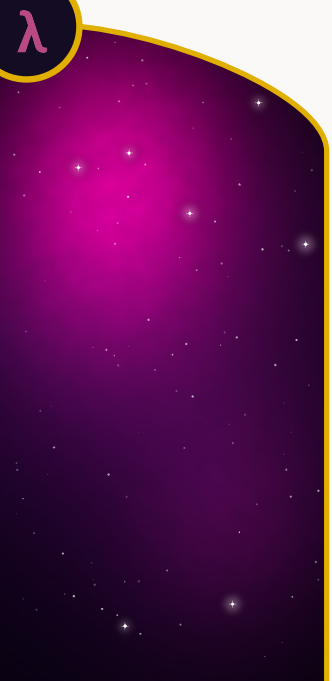
The global environment

The global environment

```
built_in_function_symbols = map(head, built_in_functions)
built_in_function_objects = map(
    lambda fun: llist('built_in', head(tail(fun))),
    built_in_functions)

def setup_environment():
    built_in_environment = extend_environment(
        append(built_in_function_symbols,
               built_in_constant_symbols),
        append(built_in_function_objects,
               built_in_constant_values),
        the_empty_environment)
    return extend_environment(None, None,
                              built_in_environment)

the_global_environment = setup_environment()
```



Evaluating applications

Evaluating applications

In `evaluate`:

```
    elif is_application(component):
        return apply(
            evaluate(function_expression(component), env),
            llist_of_values(arg_expressions(component),
                            env))
    elif is_operator_combination(component):
        app = operator_combination_to_application(
            component)
        return evaluate(app, env)
```



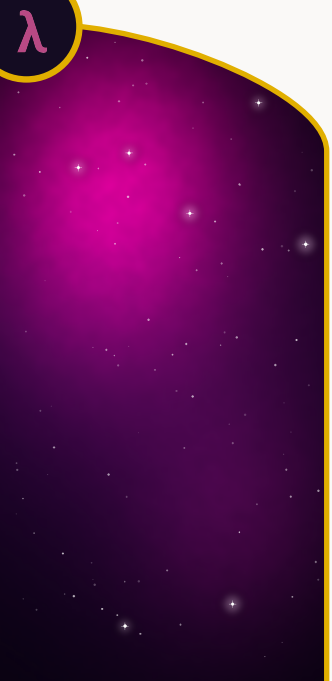
Evaluating applications

In `evaluate`:

```
    elif is_application(component):
        return apply(
            evaluate(function_expression(component), env),
            llist_of_values(arg_expressions(component),
                            env))
    elif is_operator_combination(component):
        app = operator_combination_to_application(
            component)
        return evaluate(app, env)
```

The arguments are evaluated to a linked list of values:

```
def llist_of_values(exps, env):
    return map(lambda arg: evaluate(arg, env), exps)
```



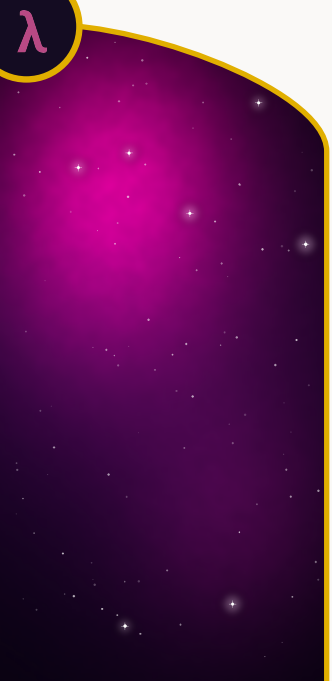
Syntax functions: applications

Syntax functions: applications

```
def is_application(component):
    return is_tagged_llist(component, 'application')

def function_expression(component):
    return head(tail(component))

def arg_expressions(component):
    return head(tail(tail(component)))
```



Built-in functions

Built-in functions

```
def is_built_in_function(fun):
    return is_tagged_llist(fun, 'built_in')

def built_in_implementation(fun):
    return head(tail(fun))

def apply_built_in_function(fun, arglist):
    return apply_in_underlying_python(
        built_in_implementation(fun), arglist)
```

Built-in functions

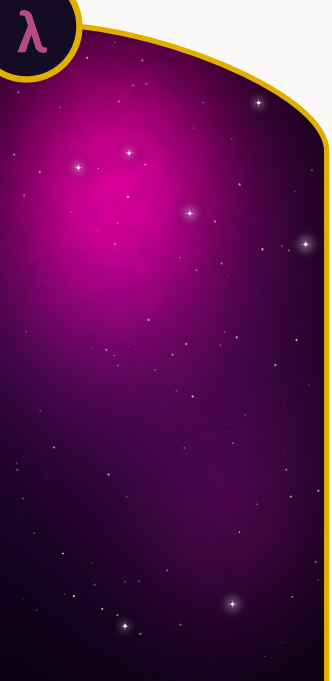
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def is_built_in_function(fun):
    return is_tagged_llist(fun, 'built_in')

def built_in_implementation(fun):
    return head(tail(fun))

def apply_built_in_function(fun, arglist):
    return apply_in_underlying_python(
        built_in_implementation(fun), arglist)
```

`apply` applies a built-in function to the arguments:

```
def apply(fun, arguments):
    if is_built_in_function(fun):
        return apply_built_in_function(fun, arguments)
    else:
        return error('unknown function type', fun)
```



Example: applying a built-in function

Example: applying a built-in function

```
tree = parse('abs(-5)')
env = the_global_environment
fun = evaluate(function_expression(tree), env)
args = llist_of_values(arg_expressions(tree), env)
print(is_built_in_function(fun))
print(apply(fun, args))
```

Example: applying a built-in function

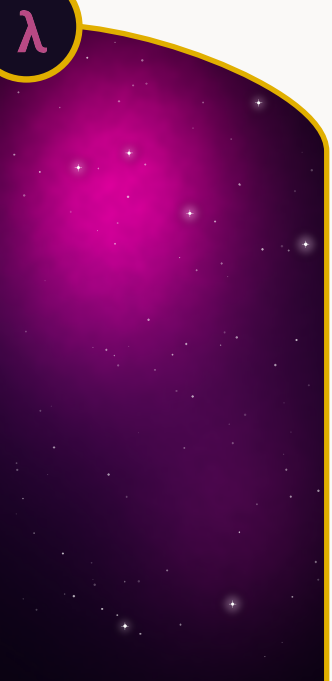
```
tree = parse('abs(-5)')
env = the_global_environment
fun = evaluate(function_expression(tree), env)
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print(is_built_in_function(fun))
print(apply(fun, args))
```

- `evaluate` finds the built-in function object of `abs` in the global environment

Example: applying a built-in function

```
tree = parse('abs(-5)')
env = the_global_environment
fun = evaluate(function_expression(tree), env)
args = llist_of_values(arg_expressions(tree), env)
print(is_built_in_function(fun))
print(apply(fun, args))
```

- `evaluate` finds the built-in function object of `abs` in the global environment
- `apply` applies Python's `abs` to the linked list of argument values



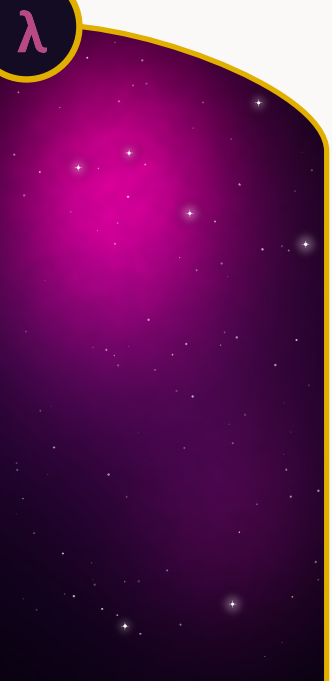
Operator combinations are applications

Operator combinations are applications

Operator combinations become applications of built-in functions:

```
def operator_combination_to_application(component):
    operator = operator_symbol(component)
    if is_unary_operator_combination(component):
        return make_application(
            make_name(operator),
            llist(first_operand(component)))
    else:
        return make_application(
            make_name(operator),
            llist(first_operand(component),
                  second_operand(component)))
```





Example: operator combinations

Example: operator combinations

```
tree = parse('1 + 2 * 3')
app = operator_combination_to_application(tree)
print(app == llist('application',
                   llist('name', '+'),
                   llist(llist('literal', 1),

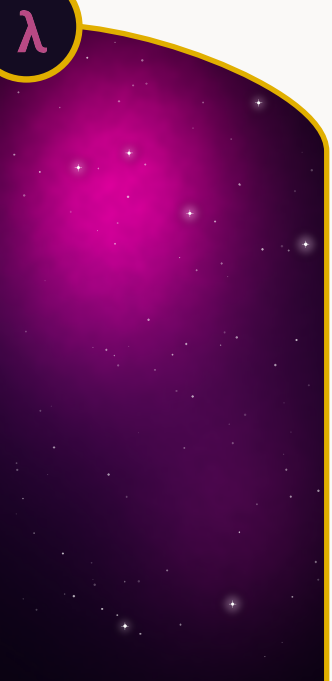
    llist('binary_operator_combination',
                                '*', llist('literal', 2),
                                llist('literal', 3)))))
print(evaluate(tree, the_global_environment))
```

Example: operator combinations

```
tree = parse('1 + 2 * 3')
app = operator_combination_to_application(tree)
print(app == llist('application',
                   llist('name', '+'),
                   llist(llist('literal', 1),

    llist('binary_operator_combination',
                                '*', llist('literal', 2),
                                llist('literal', 3)))))
print(evaluate(tree, the_global_environment))
```

Only the outer combination is transformed; `2 * 3` is transformed when `evaluate` reaches it as an argument expression.



Recap: the evaluator so far

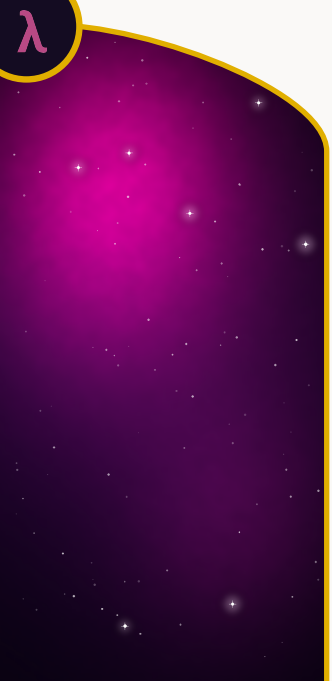
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Adding functions

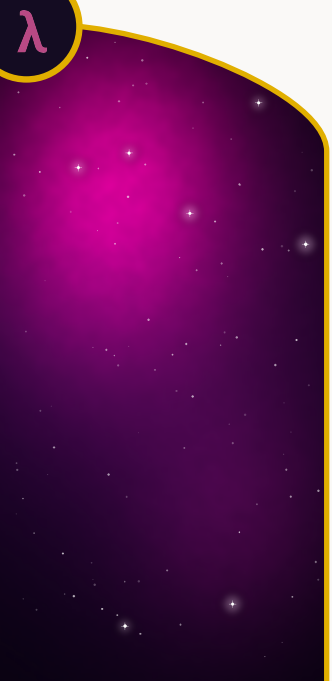
stmt	::=	$\cdots$	
	|	`def` *name* `(` *params* `):` *stmts*	function definition
	|	`return` *expr*	return statement
expr	::=	$\cdots$	
	|	*expr*`(` $expr_1$, $\ldots$, $expr_n$`)`	application
	|	`lambda` *params* `:` *expr*	lambda

Adding functions

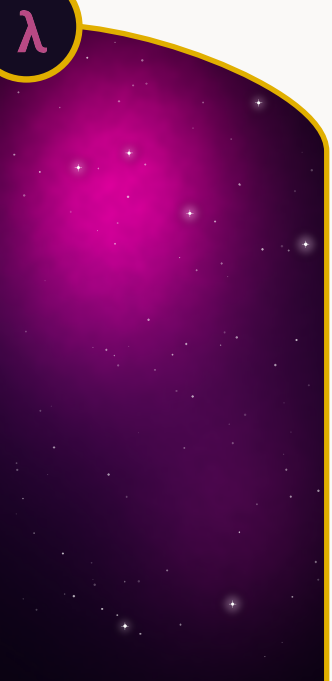
stmt	::=	⋯	
	|	`def` *name* `(` *params* `):` *stmts*	function definition
	|	`return` *expr*	return statement
expr	::=	⋯	
	|	*expr*`(` $expr_1$`, ...,` $expr_n$`)`	application
	|	`lambda` *params* `:` *expr*	lambda

Example:

```
set_evaluator(parse_and_evaluate)
set_program_text("""
def fact(n):
    return 1 if n == 1 else n * fact(n - 1)
print(fact(5))
""")
```



Parsing function definitions



Parsing function definitions

`parse('def f(x):\n    return x + 1')` returns

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`parse('def f(x):\n    return x + 1')` returns

```
tree = llist('function_definition',
             llist('name', 'f'),
             llist(llist('name', 'x')),
             llist('return_statement',
                   llist('binary_operator_combination',
                         '+',
                         llist('name', 'x'),
                         llist('literal', 1))))
print(parse('def f(x):\n    return x + 1') == tree)
```

Parsing function definitions

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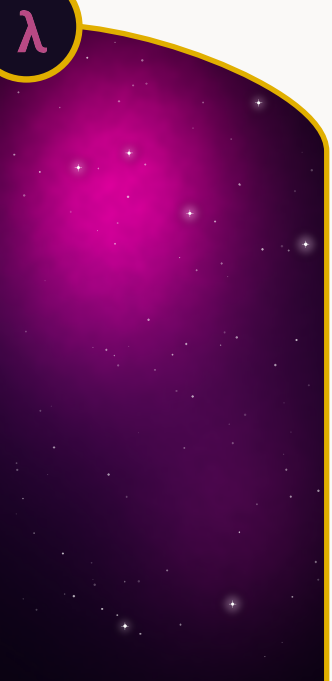
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Parsing function definitions

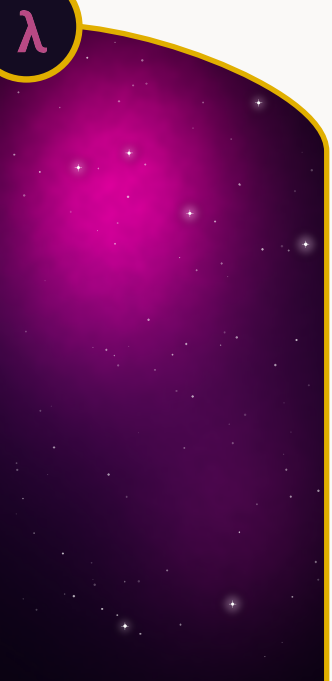
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tree = llist('function_definition',
             llist('name', 'f'),
             llist(llist('name', 'x')),
             llist('return_statement',
                   llist('binary_operator_combination',
                         '+',
                         llist('name', 'x'),
                         llist('literal', 1))))
print(parse('def f(x):\n    return x + 1') == tree)
```

- Syntax predicate: `is_function_definition`
- Selectors: `function_definition_symbol`, `function_definition_parameter_symbols`, `function_definition_body`



Parsing lambda expressions and applications



Parsing lambda expressions and applications

`parse('lambda x: x * 2')` and `parse('f(1, 2)')` return

Parsing lambda expressions and applications

`parse('lambda x: x * 2')` and `parse('f(1, 2)')` return

```
tree1 = llist('lambda_expression',
              llist(llist('name', 'x')),
              llist('binary_operator_combination',
                    '*',
                    llist('name', 'x'),
                    llist('literal', 2)))
tree2 = llist('application',
              llist('name', 'f'),
              llist(llist('literal', 1),
                    llist('literal', 2)))
print(parse('lambda x: x * 2') == tree1)
print(parse('f(1, 2)') == tree2)
```

Parsing lambda expressions and applications

`parse('lambda x: x * 2')` and `parse('f(1, 2)')` return

```
tree1 = llist('lambda_expression',
              llist(llist('name', 'x')),
              llist('binary_operator_combination',
                    '*',
                    llist('name', 'x'),
                    llist('literal', 2)))
tree2 = llist('application',
              llist('name', 'f'),
              llist(llist('literal', 1),
                    llist('literal', 2)))
print(parse('lambda x: x * 2') == tree1)
print(parse('f(1, 2)') == tree2)
```

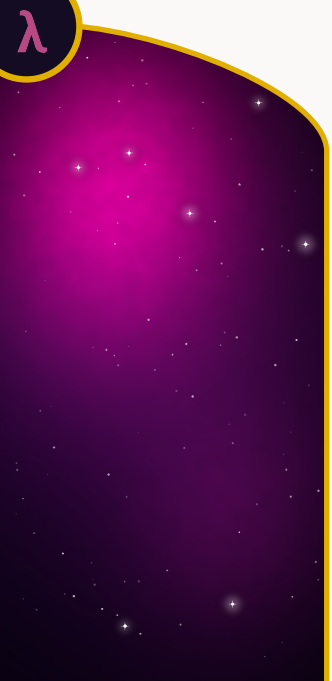
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Parsing lambda expressions and applications

`parse('lambda x: x * 2')` and `parse('f(1, 2)')` return

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tree1 = llist('lambda_expression',
              llist(llist('name', 'x')),
              llist('binary_operator_combination',
                    '*',
                    llist('name', 'x'),
                    llist('literal', 2)))
tree2 = llist('application',
              llist('name', 'f'),
              llist(llist('literal', 1),
                    llist('literal', 2)))
print(parse('lambda x: x * 2') == tree1)
print(parse('f(1, 2)') == tree2)
```

- Syntax predicate: `is_lambda_expression`
- Selectors: `lambda_parameter_symbols`, `lambda_body`



Syntax functions: function definitions

Syntax functions: function definitions

```
def is_function_definition(component):
    return is_tagged_llist(component,
    'function_definition')

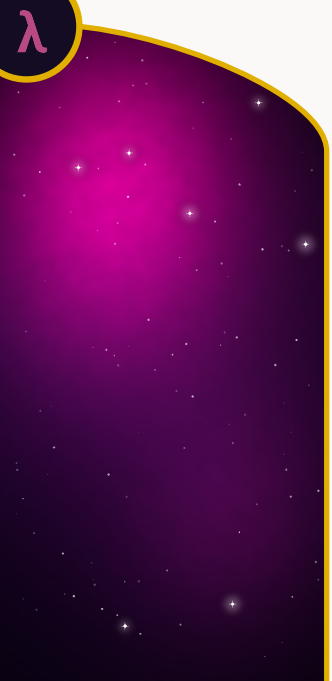
def function_definition_name(component):
    return llist_ref(component, 1)

def function_definition_symbol(component):
    return symbol_of_name(
        function_definition_name(component))

def function_definition_parameters(component):
    return llist_ref(component, 2)

def function_definition_parameter_symbols(component):
    return map(symbol_of_name,
               function_definition_parameters(component))

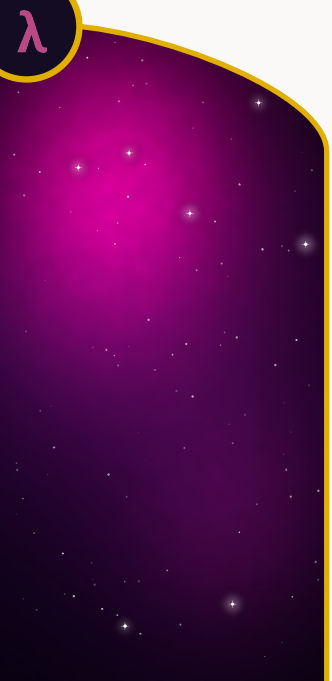
def function_definition_body(component):
    return llist_ref(component, 3)
```



Syntax functions: lambda, return

Syntax functions: lambda, return

```
def is_lambda_expression(component):
    return is_tagged_llist(component,
                           'lambda_expression')
def lambda_parameter_symbols(component):
    return map(symbol_of_name, head(tail(component)))
def lambda_body(component):
    return head(tail(tail(component)))
def is_return_statement(component):
    return is_tagged_llist(component,
                           'return_statement')
def return_expression(component):
    return head(tail(component))
def make_return_statement(expression):
    return llist('return_statement', expression)
```



Evaluating functions

Evaluating functions

In `evaluate`:

```
    elif is_lambda_expression(component):
        return make_function(
            lambda_parameter_symbols(component),
            None, None,
            make_return_statement(lambda_body(component)),
            env)
    elif is_function_definition(component):
        return eval_function_definition(component, env)
    elif is_return_statement(component):
        return eval_return_statement(component, env)
```

Evaluating functions

In `evaluate`:

```
    elif is_lambda_expression(component):
        return make_function(
            lambda_parameter_symbols(component),
            None, None,
            make_return_statement(lambda_body(component)),
            env)
    elif is_function_definition(component):
        return eval_function_definition(component, env)
    elif is_return_statement(component):
        return eval_return_statement(component, env)
```

- A lambda expression makes a function whose body is a single return statement; it cannot contain assignments or declarations, so it has no local or global names (`None`)

Evaluating functions

In `evaluate`:

```
    elif is_lambda_expression(component):
        return make_function(
            lambda_parameter_symbols(component),
            None, None,
            make_return_statement(lambda_body(component)),
            env)
    elif is_function_definition(component):
        return eval_function_definition(component, env)
    elif is_return_statement(component):
        return eval_return_statement(component, env)
```

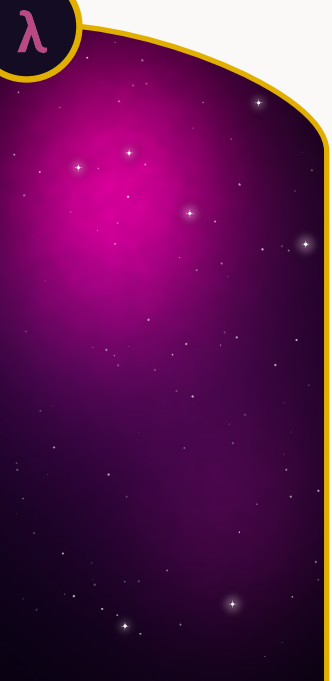
- A lambda expression makes a function whose body is a single return statement; it cannot contain assignments or declarations, so it has no local or global names (`None`)
- A function definition makes a function and assigns it to the name

Evaluating functions

In `evaluate`:

```
    elif is_lambda_expression(component):
        return make_function(
            lambda_parameter_symbols(component),
            None, None,
            make_return_statement(lambda_body(component)),
            env)
    elif is_function_definition(component):
        return eval_function_definition(component, env)
    elif is_return_statement(component):
        return eval_return_statement(component, env)
```

- A lambda expression makes a function whose body is a single return statement; it cannot contain assignments or declarations, so it has no local or global names (`None`)
- A function definition makes a function and assigns it to the name
- A return statement evaluates to a *return value*



Function objects

Function objects

```
def make_function(parameters, locals, globals,
                  body, env):
    return llist('compound_function', parameters,
                 locals, globals, body, env)

def function_parameters(f):
    return llist_ref(f, 1)
def function_locals(f):
    return llist_ref(f, 2)
def function_globals(f):
    return llist_ref(f, 3)
def function_body(f):
    return llist_ref(f, 4)
def function_environment(f):
    return llist_ref(f, 5)
```

Function objects

```
def make_function(parameters, locals, globals,
                  body, env):
    return llist('compound_function', parameters,
                 locals, globals, body, env)

def function_parameters(f):
    return llist_ref(f, 1)
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    return llist_ref(f, 2)
def function_globals(f):
    return llist_ref(f, 3)
def function_body(f):
    return llist_ref(f, 4)
def function_environment(f):
    return llist_ref(f, 5)
```

- A function object carries its local names and its global names

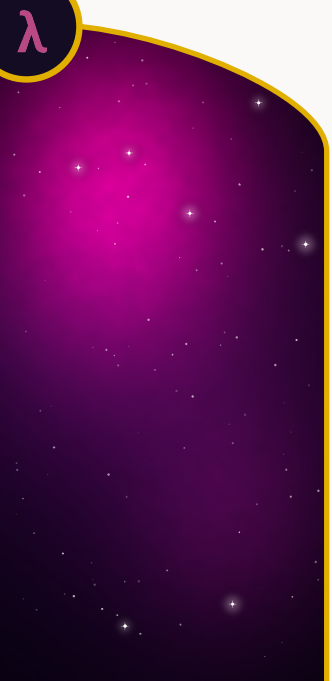
Function objects

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def function_body(f):
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def function_environment(f):
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```



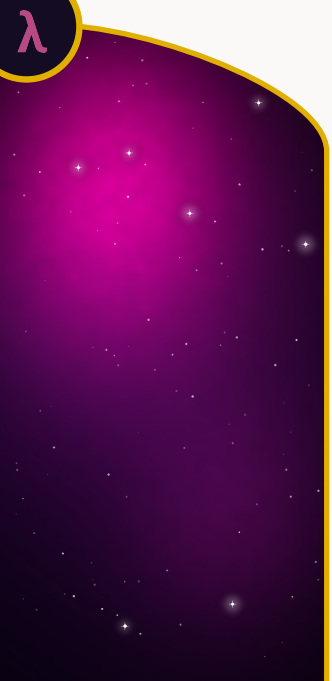
- A function object carries its local names and its global names
- A function definition determines them; see Section “Local names”



Return statements

The Problems

- Functions that do not evaluate a return statement must return `None`



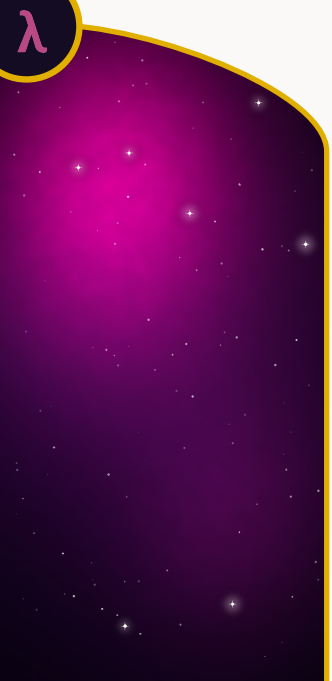
Return statements

The Problems

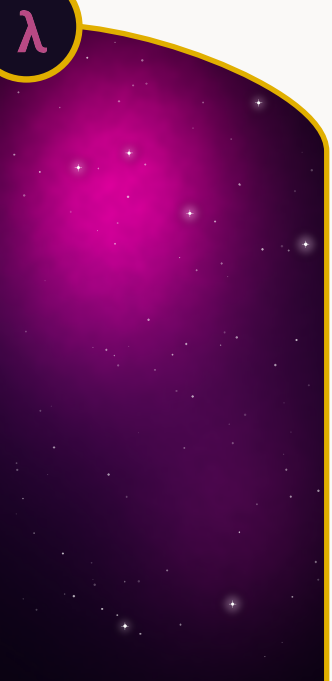
- Functions that do not evaluate a return statement must return `None`
- Evaluation of a return statement *anywhere* in the function body will return from the function with the result of evaluating the return expression

Return statements: an example

```
set_evaluator(parse_and_evaluate)
set_program_text("""
def f(n, b):
    if b:
        return n
    return -n
print(f(5, True))
print(f(5, False))
""")
```



Representing return values

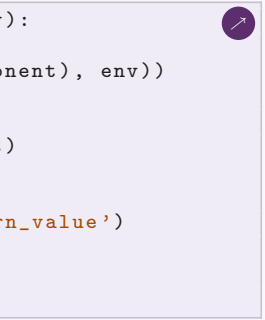


Representing return values

Evaluating a return statement wraps the value in a *return value*:

Representing return values

Evaluating a return statement wraps the value in a *return value*:

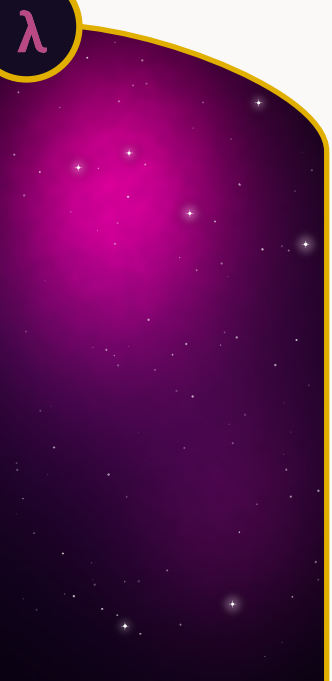


```
def eval_return_statement(component, env):
    return make_return_value(
        evaluate(return_expression(component), env))

def make_return_value(content):
    return llist('return_value', content)

def is_return_value(value):
    return is_tagged_llist(value, 'return_value')

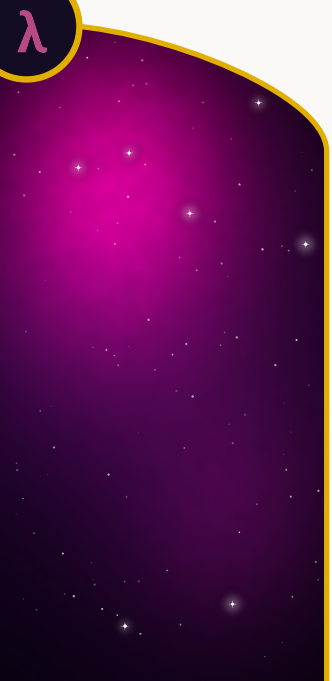
def return_value_content(value):
    return head(tail(value))
```



Return values in sequences

Return values in sequences

```
def eval_sequence(stmts, env):
    if is_empty_sequence(stmts):
        return None
    elif is_last_statement(stmts):
        return evaluate(first_statement(stmts), env)
    else:
        first_stmt_value = evaluate(
            first_statement(stmts), env)
        if is_return_value(first_stmt_value):
            return first_stmt_value
        else:
            return eval_sequence(
                rest_statements(stmts), env)
```



Applying compound functions (first try)

Applying compound functions (first try)



```
def apply(fun, arguments):
    if is_built_in_function(fun):
        return apply_built_in_function(fun, arguments)
    elif is_compound_function(fun):
        result = evaluate(
            function_body(fun),
            extend_environment(
                function_parameters(fun),
                arguments,
                function_environment(fun)))
        if is_return_value(result):
            return return_value_content(result)
        else:
            return None
    else:
        return error('unknown function type', fun)
```

Applying compound functions (first try)



```
def apply(fun, arguments):
    if is_built_in_function(fun):
        return apply_built_in_function(fun, arguments)
    elif is_compound_function(fun):
        result = evaluate(
            function_body(fun),
            extend_environment(
                function_parameters(fun),
                arguments,
                function_environment(fun)))
        if is_return_value(result):
            return return_value_content(result)
        else:
            return None
    else:
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```

- The body is evaluated in a new frame for the parameters

Applying compound functions (first try)

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                arguments,
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        if is_return_value(result):
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        else:
            return None
    else:
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```

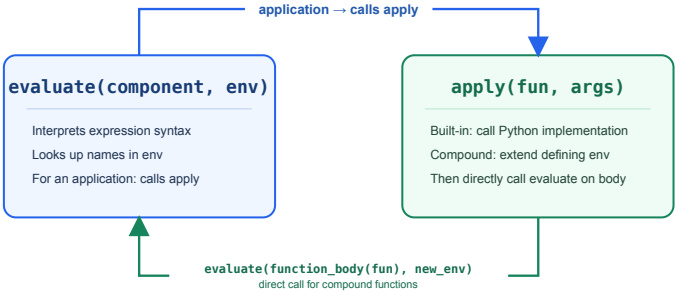
- The body is evaluated in a new frame for the parameters
- Only `apply` unwraps a return value; no `return` gives `None`

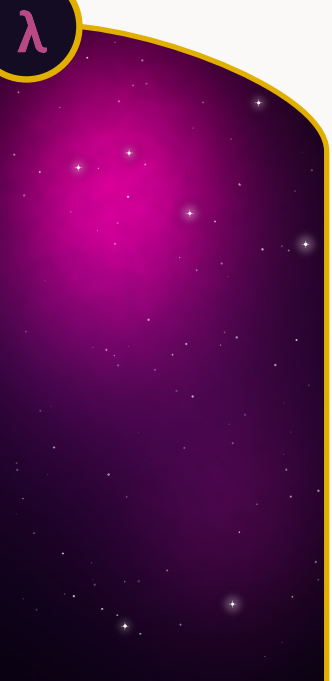
Applying compound functions (first try)

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        result = evaluate(
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            extend_environment(
                function_parameters(fun),
                arguments,
                function_environment(fun)))
        if is_return_value(result):
            return return_value_content(result)
        else:
            return None
    else:
        return error('unknown function type', fun)
```

- The body is evaluated in a new frame for the parameters
- Only `apply` unwraps a return value; no `return` gives `None`
- For now, we ignore local and global names

The evaluate–apply loop





Recap: the evaluator so far

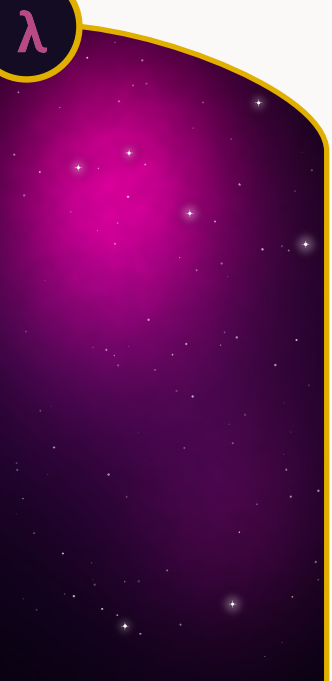
Applications and built-in functions

Adding functions

Local names

Running the evaluator

Lazy evaluation



Names in function bodies

Names in function bodies

```
set_evaluator(parse_and_evaluate)
set_program_text("""
x = 1
def g():
    x = 2
    return x
print(g())
print(x)
""")
```



Names in function bodies

```
set_evaluator(parse_and_evaluate)
set_program_text("""
x = 1
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print(g())
print(x)
""")
```



- An assignment in a function body makes the name *local* to the function

Names in function bodies

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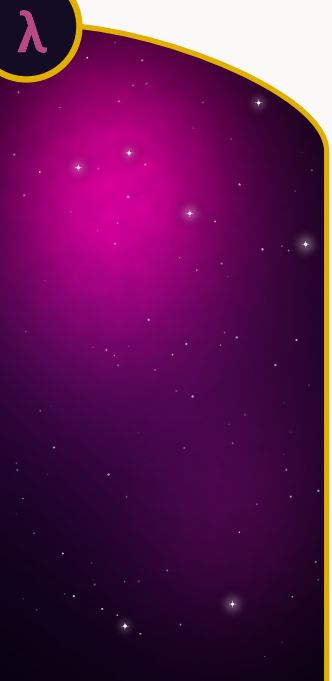
- An assignment in a function body makes the name *local* to the function
- `global x`: the name refers to the global frame

Names in function bodies

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set_evaluator(parse_and_evaluate)
set_program_text("""
x = 1
def g():
    x = 2
    return x
print(g())
print(x)
""")
```



- An assignment in a function body makes the name *local* to the function
- `global x`: the name refers to the global frame
- `nonlocal x`: the name refers to an enclosing function's frame



Function definitions

Function definitions

```
def eval_function_definition(component, env):
    parameters = function_definition_parameter_symbols(
        component)
    body = function_definition_body(component)
    fun = make_function(parameters,
                        local_variables(parameters, body),
                        global_variables(body),
                        body, env)
    assign_symbol_value(
        function_definition_symbol(component), fun, env)
    return None
def local_variables(parameters, body):
    excluded = append(parameters,
                      append(global_variables(body),
                             nonlocal_variables(body)))
    return filter(
        lambda s: is_none(member(s, excluded)),
        assigned_variables(body))
```

Function definitions

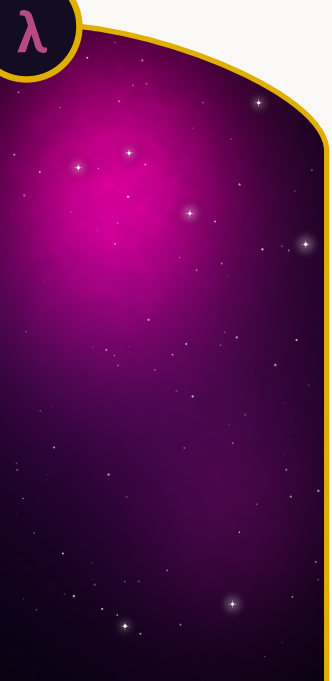
```
def eval_function_definition(component, env):
    parameters = function_definition_parameter_symbols(
        component)
    body = function_definition_body(component)
    fun = make_function(parameters,
                        local_variables(parameters, body),
                        global_variables(body),
                        body, env)
    assign_symbol_value(
        function_definition_symbol(component), fun, env)
    return None
def local_variables(parameters, body):
    excluded = append(parameters,
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    return filter(
        lambda s: is_none(member(s, excluded)),
        assigned_variables(body))
```

- Global names: declared by `global` statements in the body

Function definitions

```
def eval_function_definition(component, env):
    parameters = function_definition_parameter_symbols(
        component)
    body = function_definition_body(component)
    fun = make_function(parameters,
                        local_variables(parameters, body),
                        global_variables(body),
                        body, env)
    assign_symbol_value(
        function_definition_symbol(component), fun, env)
    return None
def local_variables(parameters, body):
    excluded = append(parameters,
                      append(global_variables(body),
                             nonlocal_variables(body)))
    return filter(
        lambda s: is_none(member(s, excluded)),
        assigned_variables(body))
```

- Global names: declared by `global` statements in the body
- Local names: assigned in body (not `global`/`nonlocal`)



Applying compound functions (complete)

Applying compound functions (complete)

```
def apply(fun, arguments):
    if is_built_in_function(fun):
        return apply_built_in_function(fun, arguments)
    elif is_compound_function(fun):
        locals = function_locals(fun)
        globals = function_globals(fun)
        unassigned_markers = map(
            lambda s: '*unassigned*', locals)
        global_markers = map(
            lambda s: '*global*', globals)
```

Applying compound functions (complete)

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def apply(fun, arguments):
    if is_built_in_function(fun):
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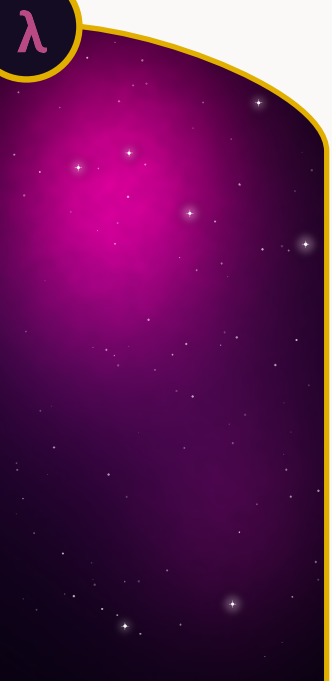
- The function object supplies the local and global names

Applying compound functions (complete)



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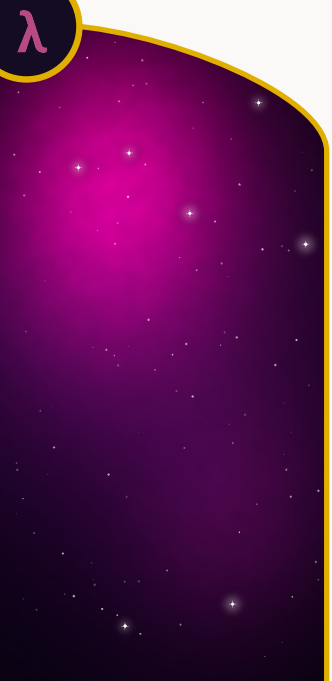
- The function object supplies the local and global names
- Local names start out `'*unassigned*'`; global names are marked `'*global*'`



Applying compound functions (cont.)

Applying compound functions (cont.)

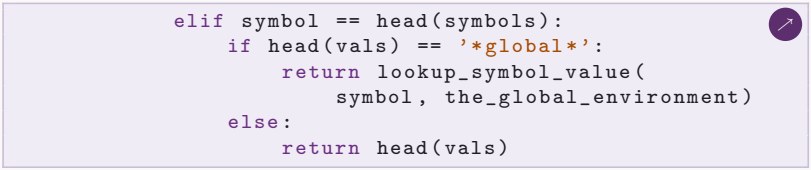
```
        new_env = extend_environment(
            append(function_parameters(fun),
                   append(locals, globals)),
            append(arguments,
                   append(unassigned_markers,
                          global_markers)),
            function_environment(fun))
        result = evaluate(function_body(fun), new_env)
        if is_return_value(result):
            return return_value_content(result)
        else:
            return None
    else:
        return error('unknown function type', fun)
```



Names marked `*global*`

Names marked *global*

In lookup_symbol_value:



```
        elif symbol == head(symbols):
            if head(vals) == '*global*':
                return lookup_symbol_value(
                    symbol, the_global_environment)
            else:
                return head(vals)
```

Names marked *global*

In lookup_symbol_value:

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            elif symbol == head(symbols):
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                    return lookup_symbol_value(
                        symbol, the_global_environment)
                else:
                    return head(vals)
```

In assign_symbol_value:

```
        elif symbol == head(symbols):
            if head(vals) == '*global*':
                assign_symbol_value(
                    symbol, val, the_global_environment)
            else:
                set_head(vals, val)
```

Names marked *global*

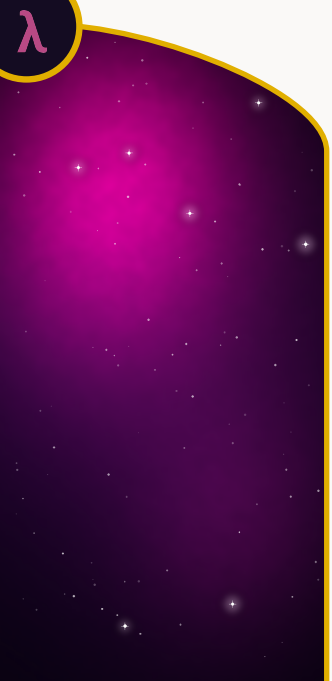
In lookup_symbol_value:

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                        symbol, the_global_environment)
                else:
                    return head(vals)
```

In assign_symbol_value:

```
        elif symbol == head(symbols):
            if head(vals) == '*global*':
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                    symbol, val, the_global_environment)
            else:
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```

nonlocal names are simply not made local, so that the search continues in the enclosing environment.



Recap: the evaluator so far

Applications and built-in functions

Adding functions

Local names

Running the evaluator

Lazy evaluation

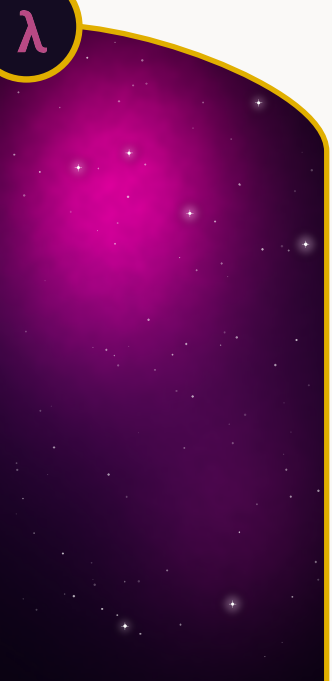
Running the evaluator

```
from repl import set_evaluator, set_program_text

def parse_and_evaluate(program):
    evaluate(parse(program), the_global_environment)

set_evaluator(parse_and_evaluate)
set_program_text("""
def f(n, b):
    if b:
        return n
    return -n
x = 1
def g():
    x = 2
    return x
print(f(5, False))
print(g())
print(x)
""")
```

The Repl tab calls `parse_and_evaluate` with every program that you run there; programs print with `print`.



Recap: the evaluator so far

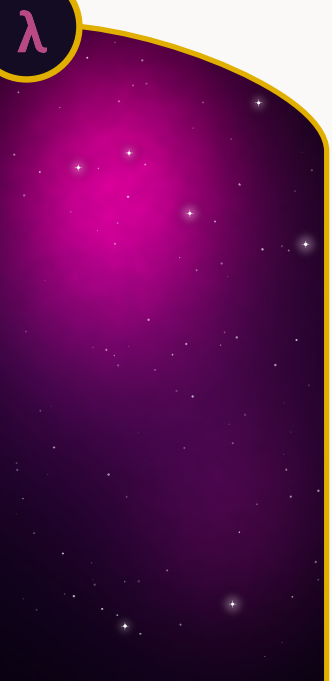
Applications and built-in functions

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An example

An example

```
set_evaluator(parse_and_evaluate)
set_program_text("""
def unless(condition, usual, exceptional):
    return exceptional if condition else usual
xs = None
unless(is_none(xs), head(xs), print("xs is empty"))
""")
```

An example

```
set_evaluator(parse_and_evaluate)
set_program_text("""
def unless(condition, usual, exceptional):
    return exceptional if condition else usual
xs = None
unless(is_none(xs), head(xs), print("xs is empty"))
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```

Can we implement `unless` like this? No: Python evaluates all three arguments *before* calling `unless`, so `head(xs)` fails when `xs` is empty.

An example

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set_program_text("""
def unless(condition, usual, exceptional):
    return exceptional if condition else usual
xs = None
unless(is_none(xs), head(xs), print("xs is empty"))
""")
```

Can we implement `unless` like this? No: Python evaluates all three arguments *before* calling `unless`, so `head(xs)` fails when `xs` is empty.

Open question

Python evaluates function arguments before calling the function. Must a language work this way?

Recall: applicative and normal order (Lecture L2A)

Applicative order:

```
->  sum_of_squares(5 + 1, 5 * 2)
->  sum_of_squares(6, 10)
->  square(6) + square(10)
```

Recall: applicative and normal order (Lecture L2A)

Applicative order:

```
->  sum_of_squares(5 + 1, 5 * 2)
->  sum_of_squares(6, 10)
->  square(6) + square(10)
```

Normal order:

```
->  sum_of_squares(5 + 1, 5 * 2)
->  square(5 + 1) + square(5 * 2)
->  ((5 + 1) * (5 + 1)) + ((5 * 2) * (5 * 2))
```

Recall: applicative and normal order (Lecture L2A)

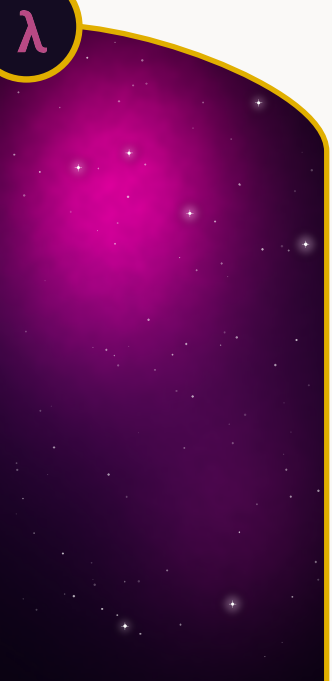
Applicative order:

```
->  sum_of_squares(5 + 1, 5 * 2)
->  sum_of_squares(6, 10)
->  square(6) + square(10)
```

Normal order:

```
->  sum_of_squares(5 + 1, 5 * 2)
->  square(5 + 1) + square(5 * 2)
->  ((5 + 1) * (5 + 1)) + ((5 * 2) * (5 * 2))
```

Python, like our evaluator so far, uses applicative order.



Lazy evaluation

Laziness

In applications, arguments are not evaluated. They are passed to the function as expressions.

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Force evaluation when needed

The argument expressions are evaluated when needed, during evaluation of the body

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In applications, arguments are not evaluated. They are passed to the function as expressions.

Force evaluation when needed

The argument expressions are evaluated when needed, during evaluation of the body

Non-strict and strict

Compound functions are non-strict in each argument; built-in functions stay strict.

Implementation

"Eager" evaluation (applicative order)

```
    elif is_application(component):
        return apply(
            evaluate(function_expression(component), env),
            llist_of_values(arg_expressions(component),
                            env))
```

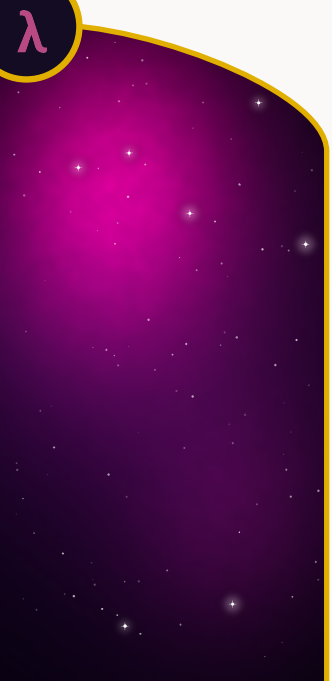
Implementation

"Eager" evaluation (applicative order)

```
    elif is_application(component):
        return apply(
            evaluate(function_expression(component), env),
            llist_of_values(arg_expressions(component),
                            env))
```

"Lazy" evaluation (normal order)

```
    elif is_application(component):
        return apply(
            actual_value(
                function_expression(component), env),
            arg_expressions(component),
            env)
```



Modifying `apply`

Modifying `apply`

```
def apply(fun, args, env):
    if is_built_in_function(fun):
        return apply_built_in_function(
            fun, llist_of_arg_values(args, env))
    elif is_compound_function(fun):
        delayed_args = llist_of_delayed_args(args, env)
```

Modifying `apply`

```
def apply(fun, args, env):
    if is_built_in_function(fun):
        return apply_built_in_function(
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```



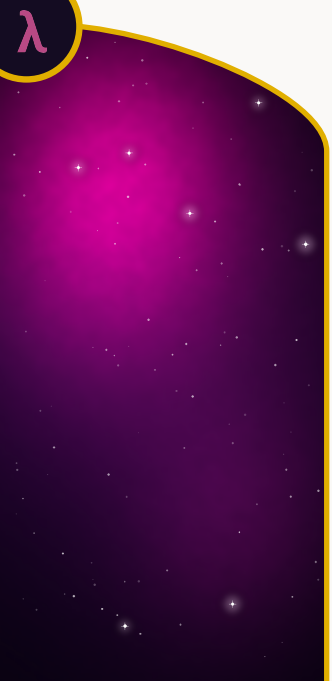
- Built-in functions are strict: their arguments are evaluated

Modifying `apply`

```
def apply(fun, args, env):
    if is_built_in_function(fun):
        return apply_built_in_function(
            fun, llist_of_arg_values(args, env))
    elif is_compound_function(fun):
        delayed_args = llist_of_delayed_args(args, env)
```



- Built-in functions are strict: their arguments are evaluated
- Compound functions are non-strict: their arguments are delayed



Argument evaluation

Argument evaluation

```
def actual_value(exp, env):
    return force_it(evaluate(exp, env))

def llist_of_arg_values(exps, env):
    return map(lambda exp: actual_value(exp, env), exps)

def llist_of_delayed_args(exps, env):
    return map(lambda exp: delay_it(exp, env), exps)
```

Argument evaluation

```
def actual_value(exp, env):
    return force_it(evaluate(exp, env))

def llist_of_arg_values(exps, env):
    return map(lambda exp: actual_value(exp, env), exps)

def llist_of_delayed_args(exps, env):
    return map(lambda exp: delay_it(exp, env), exps)
```

A conditional needs the actual value of its predicate:

```
def eval_conditional(component, env):
    predicate = conditional_predicate(component)
    if is_truthy(actual_value(predicate, env)):
        return evaluate(
            conditional_consequent(component), env)
    else:
        return evaluate(
            conditional_alternative(component), env)
```

Implementation of thunks

```
def delay_it(exp, env):
    return llist('thunk', exp, env)

def is_thunk(obj):
    return is_tagged_llist(obj, 'thunk')

def thunk_exp(thunk):
    return head(tail(thunk))

def thunk_env(thunk):
    return head(tail(tail(thunk)))

def is_evaluated_thunk(obj):
    return is_tagged_llist(obj, 'evaluated_thunk')

def thunk_value(evaluated_thunk):
    return head(tail(evaluated_thunk))
```

Implementation of forcing (simple version)

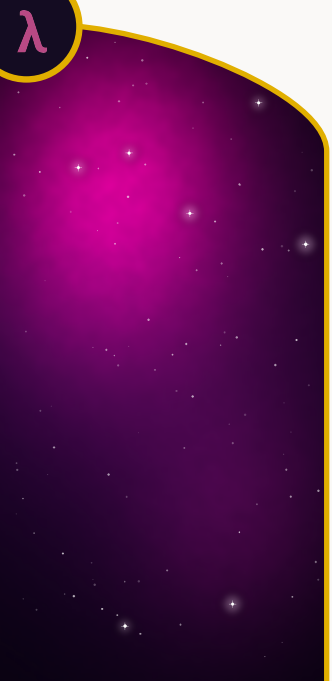


```
def force_it(obj):
    if is_thunk(obj):
        return actual_value(thunk_exp(obj),
                            thunk_env(obj))
    else:
        return obj
```

Implementation of forcing (with memoization)



```
def force_it(obj):
    if is_thunk(obj):
        result = actual_value(thunk_exp(obj),
                              thunk_env(obj))
        set_head(obj, 'evaluated_thunk')
        set_head(tail(obj), result)  # store value
        set_tail(tail(obj), None)    # drop env
        return result
    elif is_evaluated_thunk(obj):
        return thunk_value(obj)
    else:
        return obj
```



Built-in functions stay strict

Built-in functions stay strict



```
set_evaluator(parse_and_evaluate)
set_program_text("""
def make_ints(n):
    return pair(n, make_ints(n + 1))
ints = make_ints(1)
""")
```

Built-in functions stay strict

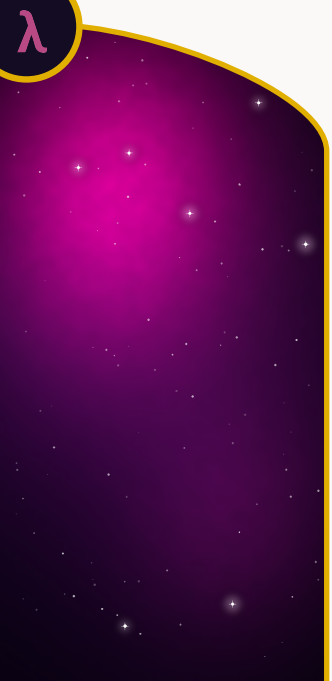
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- `make_ints` is compound, but `pair` is built-in

Built-in functions stay strict

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set_evaluator(parse_and_evaluate)
set_program_text("""
def make_ints(n):
    return pair(n, make_ints(n + 1))
ints = make_ints(1)
""")
```

- `make_ints` is compound, but `pair` is built-in
- `pair` evaluates its arguments: `make_ints(1)` needs `make_ints(2)`, which needs `make_ints(3)`, ...



Lazy pairs

Lazy pairs

```
set_evaluator(parse_and_evaluate)
set_program_text("""
def pair(x, y):
    return lambda m: m(x, y)
def head(z):
    return z(lambda p, q: p)
def tail(z):
    return z(lambda p, q: q)
def make_ints(n):
    return pair(n, make_ints(n + 1))
ints = make_ints(1)
print(head(tail(ints)))
""")
```

Lazy pairs

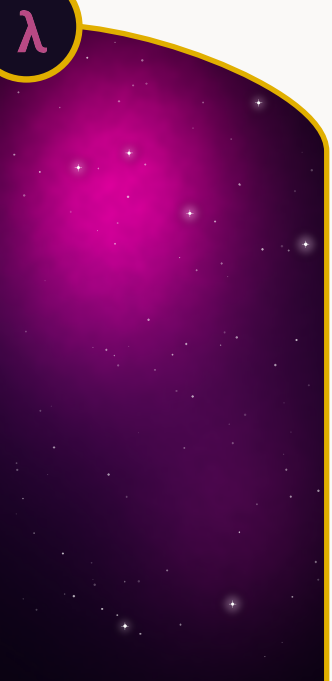
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def make_ints(n):
    return pair(n, make_ints(n + 1))
ints = make_ints(1)
print(head(tail(ints)))
""")
```

- Compound `pair`, `head` and `tail` shadow the built-in ones

Lazy pairs

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def pair(x, y):
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def make_ints(n):
    return pair(n, make_ints(n + 1))
ints = make_ints(1)
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""")
```

- Compound `pair`, `head` and `tail` shadow the built-in ones
- Compound functions are non-strict, so these pairs are lazy: infinite linked lists for free



Running the lazy evaluator

Running the lazy evaluator

```
from repl import set_evaluator, set_program_text

def parse_and_evaluate(program):
    actual_value(parse(program), the_global_environment)

set_evaluator(parse_and_evaluate)
set_program_text("""
def unless(condition, usual, exceptional):
    return exceptional if condition else usual
xs = None
unless(is_none(xs), head(xs), print("xs is empty"))
""")
```

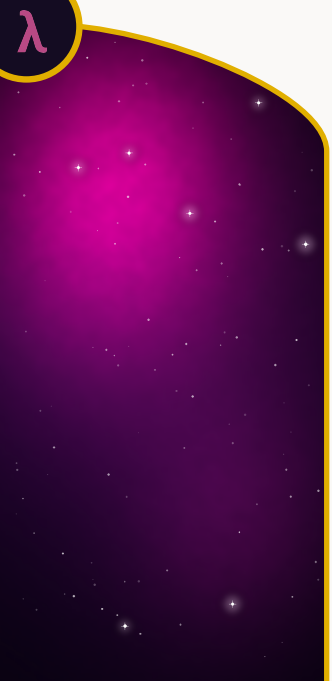
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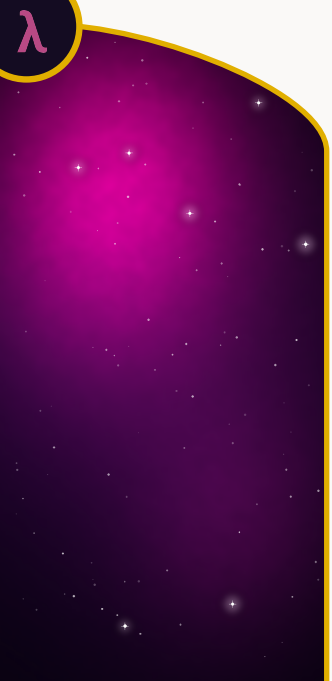
def parse_and_evaluate(program):
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set_evaluator(parse_and_evaluate)
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def unless(condition, usual, exceptional):
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xs = None
unless(is_none(xs), head(xs), print("xs is empty"))
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```

`head(xs)` is never evaluated: the argument `usual` is never forced.

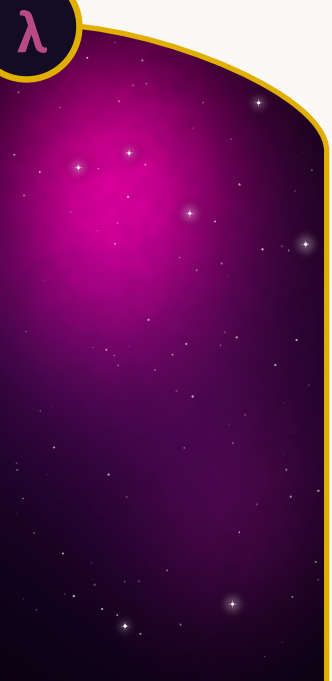


Summary



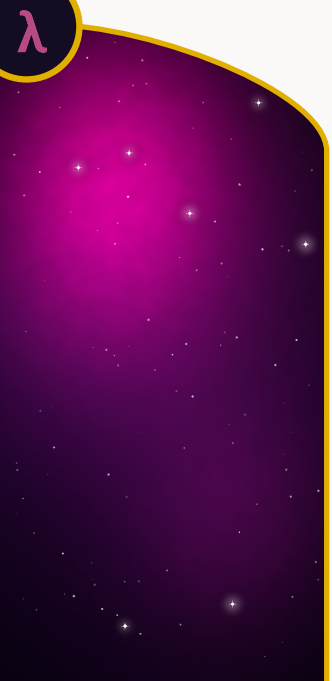
Summary

- Metacircular evaluator:
  the most beautiful program ever written (?)



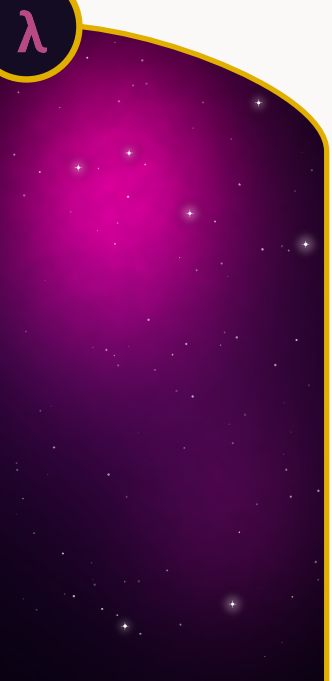
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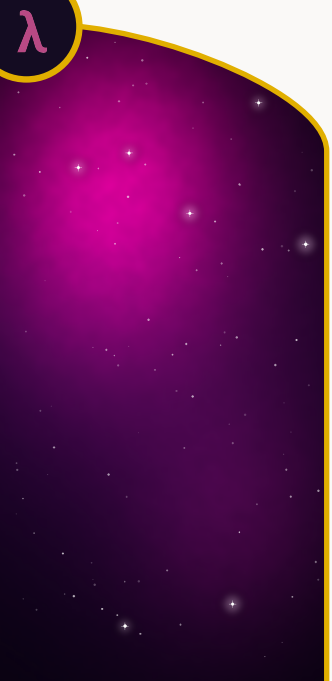
Summary

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- `evaluate`—`apply` loop



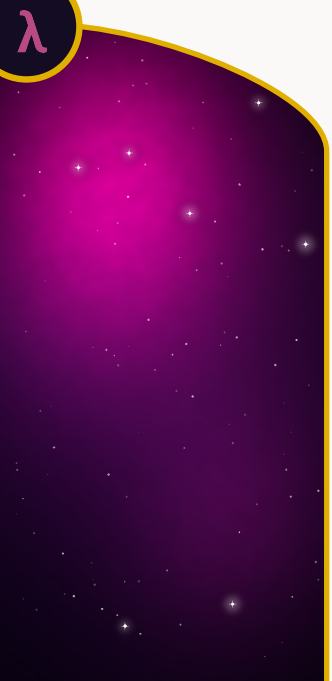
Summary

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- `evaluate`—`apply` loop
- Alternative understanding of `return`



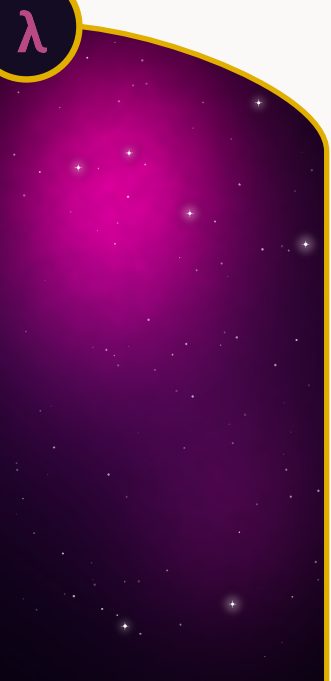
Summary

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- Local and global names: determined by function definitions, marked by `apply` in the new frame



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Summary

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- Explains the environment model
- `evaluate`—`apply` loop
- Alternative understanding of `return`
- Local and global names: determined by function definitions, marked by `apply` in the new frame
- Lazy evaluation: delay arguments as thunks, force them when needed
- What other rules of a programming language could we change?

Check-In time!

Don't be lazy, but do think about it—or should we say, *thunk* about it!

Look for Check-In I12A in the Source Academy!